National Institute Of Technology Calicut
Department of Computer Science and Engineering
CS4043D Image Processing
Assignment 1

Date of Submission : On or before 16-03-2026
Maximum Marks : 15

Instructions to the students:

- *Any submitted work should be your own. Academic dishonesty in any form can lead to zero marks for the assignment.*
- *Any work submitted after the submission deadline will not be considered for evaluation (exception may be given only for genuine reasons).*
- *Students are free to use the language/tool of their choice.(Preferably Matlab/Octave/Java/Python etc.)*
- *Prepare a PDF document that contains the question, your code and output. Save it with file name as Your Name_Rollno and upload it in the submission link given in eduserver on or before the deadline.*

1. Take a real image(Your own photograph of size 256x256) and do the following
   a) Read it to memory from the file and display it.
   b) Read a portion of it to memory and isplay it
   c) add a constant to this portion and then display it
   d) display the whole image after adding a constant to a portion of it. Take care of overflow while adding. (on overflow, take the pixel vale as the maximum possible)
   e) multiply a portion of the image by a constant ranging from 0.1 to 2.0, truncating to maximum value on over flow. Display the resulting image for each value of the constant multiplier.
   f)Create a second image which contains only your name and date of doing the assignment, and embed this to to your photograph as a visible watermark.
   g) Embed the second image(name &date) as an invisible watermark in your photograph. (also write code for extracting the watermark)
   h) Apply thresholding on the image to create a binary image.

2. Apply filtering operations on images:
Enhance the image by,
   i) Low pass filtering using 3x3 and 5x5 masks
   ii) High pass filtering using a) Sobel operator,b) Laplacian operator.
   iii) Add Gaussian noise to the image and apply median filtering to remove the noise.
   iv) Do unsharp masking .
   (Please write your own code for doing convolution in the spatial domain)

3.i) Apply Histogram Equalization in the image and display the histogram of original image, equalized image and histogram of the equalized image.
   ii) Apply contrast streching on the original image and display the output.